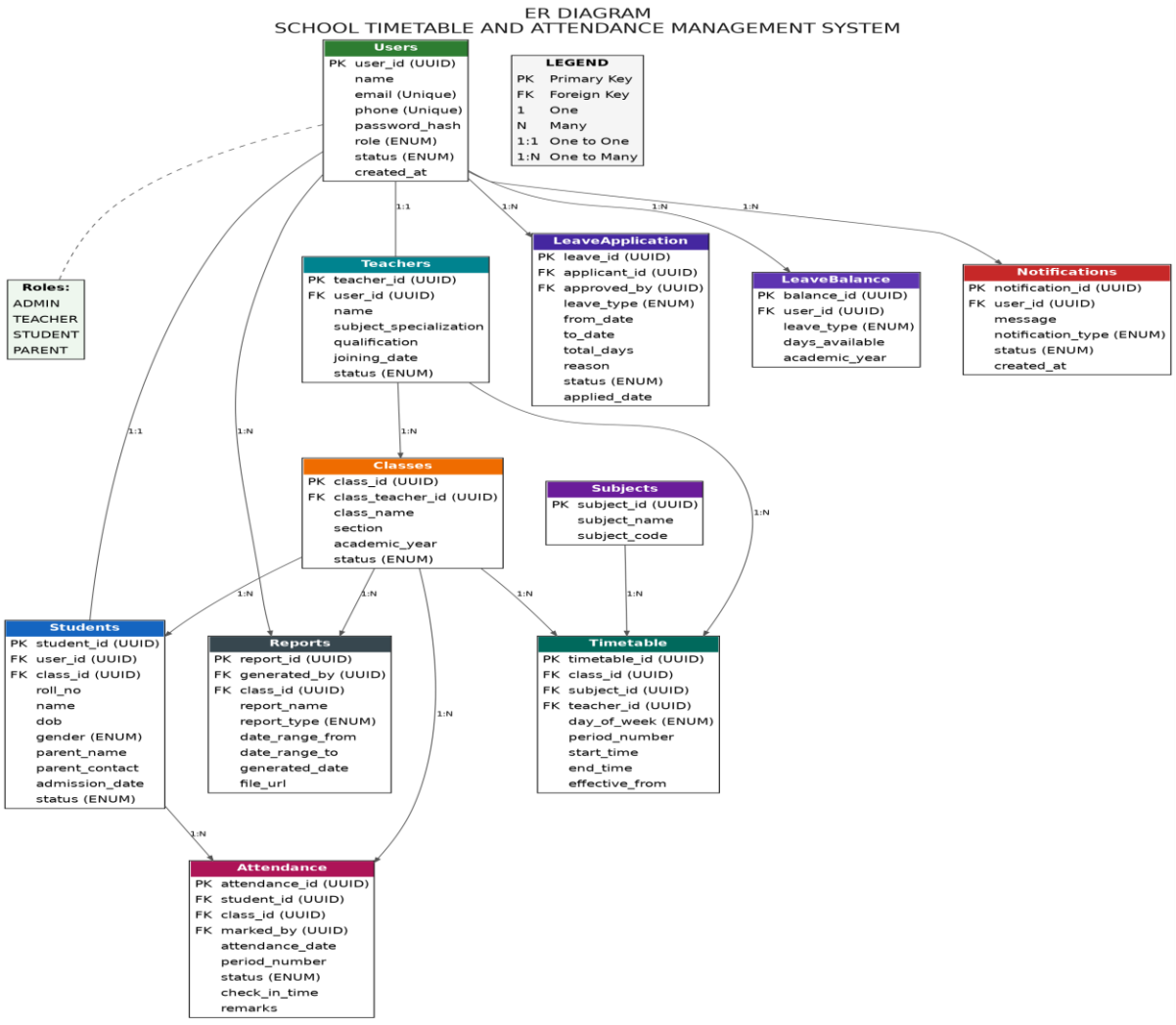
DATABASE DESIGN

SCHOOL TIMETABLE AND ATTENDANCE MANAGEMENT SYSTEM

1. Entity-Relationship Diagram

The ER diagram is centered around the Users table, which stores all system users (Admin, Teacher, Student, Parent). Users are linked to Students and Teachers, who participate in class assignment, timetable scheduling, attendance marking, leave management, and notifications.

The core business flow begins with Classes, which group Students and are assigned a Timetable built from Subjects and Teachers. Attendance is recorded per Student, per Class, per period, based on the active Timetable. Leave Applications and Leave Balances track staff and student leave, while Reports and Notifications support administrative oversight. Every table uses primary keys for unique identification and foreign keys to maintain referential integrity.



2. Table Structures

2.1 Users

Stores login information for every system user.

Field	Type	Notes
user_id	UUID	PK
name	VARCHAR(100)	Alphabetic only
email	VARCHAR(100)	UNIQUE
phone	VARCHAR(10)	UNIQUE
password_hash	VARCHAR(255)	Encrypted Password
role	ENUM	ADMIN, TEACHER, STUDENT, PARENT
status	ENUM	ACTIVE, INACTIVE
created_at	TIMESTAMP	

2.2 Students

Stores registered student profile and academic details.

Field	Type	Notes
student_id	UUID	PK
user_id	UUID	FK -> Users (nullable, if student has a login)
class_id	UUID	FK -> Classes
roll_no	VARCHAR(10)	
name	VARCHAR(100)	
dob	DATE	
gender	ENUM	MALE, FEMALE, OTHER
parent_name	VARCHAR(100)	
parent_contact	VARCHAR(10)	
address	VARCHAR(255)	
admission_date	DATE	
status	ENUM	ACTIVE, INACTIVE, TRANSFERRED

2.3 Teachers

Stores teaching staff profile and specialization.

Field	Type	Notes
teacher_id	UUID	PK
user_id	UUID	FK -> Users
name	VARCHAR(100)	
subject_specialization	VARCHAR(100)	
qualification	VARCHAR(100)	

Field	Type	Notes
joining_date	DATE	
status	ENUM	ACTIVE, INACTIVE

2.4 Classes

Stores class and section information.

Field	Type	Notes
class_id	UUID	PK
class_teacher_id	UUID	FK -> Teachers
class_name	VARCHAR(20)	e.g. X, IX, VIII
section	VARCHAR(5)	e.g. A, B, C
academic_year	VARCHAR(20)	
status	ENUM	ACTIVE, INACTIVE

2.5 Subjects

Master list of subjects taught across classes.

Field	Type	Notes
subject_id	UUID	PK
subject_name	VARCHAR(100)	e.g. Maths, Science, English
subject_code	VARCHAR(20)	

2.6 Timetable

Stores the weekly period-wise schedule for each class.

Field	Type	Notes
timetable_id	UUID	PK
class_id	UUID	FK -> Classes
subject_id	UUID	FK -> Subjects
teacher_id	UUID	FK -> Teachers
day_of_week	ENUM	MONDAY ... SATURDAY
period_number	INT	1 - 8
start_time	TIME	
end_time	TIME	
effective_from	DATE	

2.7 Attendance

Stores daily, period-wise attendance for each student.

Field	Type	Notes
attendance_id	UUID	PK
student_id	UUID	FK -> Students
class_id	UUID	FK -> Classes
marked_by	UUID	FK -> Users (teacher/admin who marked it)
attendance_date	DATE	
period_number	INT	
status	ENUM	PRESENT, ABSENT, LATE
check_in_time	TIME	
remarks	VARCHAR(255)	e.g. Late entry, On leave

2.8 LeaveApplication

Stores leave requests raised by staff or on behalf of students.

Field	Type	Notes
leave_id	UUID	PK
applicant_id	UUID	FK -> Users
approved_by	UUID	FK -> Users (nullable until reviewed)
leave_type	ENUM	CASUAL, MEDICAL, EARNED, COMP_OFF
leave_for	ENUM	SELF, ON_BEHALF
from_date	DATE	
to_date	DATE	
total_days	INT	
reason	TEXT	Minimum 10 characters
contact_number	VARCHAR(10)	
address_during_leave	VARCHAR(255)	
document_url	VARCHAR(255)	Optional supporting document
status	ENUM	PENDING, APPROVED, REJECTED
applied_date	DATE	

2.9 LeaveBalance

Tracks available leave days by type for each user.

Field	Type	Notes
balance_id	UUID	PK
user_id	UUID	FK -> Users
leave_type	ENUM	CASUAL, MEDICAL, EARNED, COMP_OFF
days_available	DECIMAL(4,1)	
academic_year	VARCHAR(20)	

2.10 Notifications

Stores system notifications sent to users.

Field	Type	Notes
notification_id	UUID	PK
user_id	UUID	FK -> Users
message	TEXT	
notification_type	ENUM	EMAIL, SMS, IN_APP
status	ENUM	READ, UNREAD
created_at	TIMESTAMP	

2.11 Reports

Stores generated attendance, leave, and performance reports.

Field	Type	Notes
report_id	UUID	PK
generated_by	UUID	FK -> Users
class_id	UUID	FK -> Classes (nullable, for school-wide reports)
report_name	VARCHAR(150)	
report_type	ENUM	ATTENDANCE, LEAVE, CLASS_WISE, STUDENT_PERFORMANCE
date_range_from	DATE	
date_range_to	DATE	
generated_date	DATE	
file_url	VARCHAR(255)	

3. Relationships

The School Timetable and Attendance Management System consists of multiple interconnected tables that manage user accounts, class and subject setup, timetable scheduling, attendance tracking, leave management, notifications, and reporting. The Users table serves as the central entity, extended by Students and Teachers profiles. Classes group Students and drive the Timetable, which in turn governs how Attendance is recorded. Leave Applications and Leave Balances are tied to Users, while Notifications and Reports are generated by or for authorized users. All tables are connected using Primary Keys (PK) and Foreign Keys (FK) to ensure data integrity and consistency throughout the system.

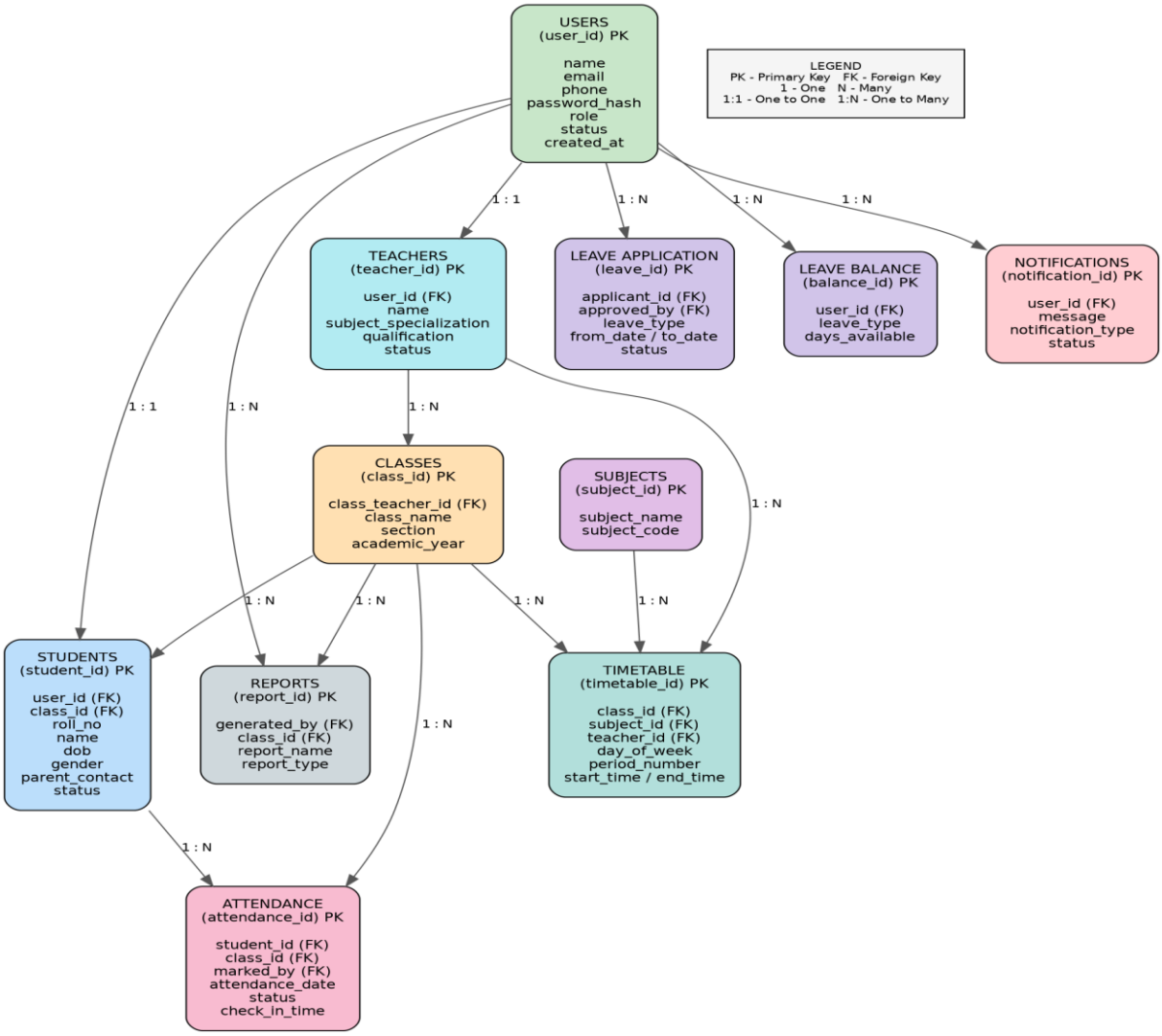
Important Constraints

- Each User has one Student profile or one Teacher profile, depending on role.
- Each Class belongs to one Class Teacher and contains multiple Students.
- Each Class has multiple Timetable entries, one per subject/period/day.
- Each Student can have multiple Attendance records, one per period per day.
- Each User (staff or student) can raise multiple Leave Applications and holds a Leave Balance per leave type.
- Notifications and Reports are linked to the authorized Users who receive or generate them.
- Primary Key (PK) and Foreign Key (FK) constraints ensure data integrity and consistency.

Table Relationships

Parent Table	Child Table	Cardinality	Description
Users	Students	1 : 1	One user has one student profile (if the student logs in).
Users	Teachers	1 : 1	One user has one teacher profile.
Classes	Students	1 : N	One class contains multiple students.
Teachers	Classes	1 : N	One teacher can be the class teacher for multiple classes.
Classes	Timetable	1 : N	One class has multiple timetable entries across the week.
Teachers	Timetable	1 : N	One teacher can be assigned multiple periods.
Subjects	Timetable	1 : N	One subject can appear in multiple timetable slots.
Students	Attendance	1 : N	One student has multiple attendance records.
Classes	Attendance	1 : N	One class has attendance records for all its students.
Users	LeaveApplication	1 : N	One user can submit multiple leave applications.
Users	LeaveBalance	1 : N	One user holds a leave balance per leave type.
Users	Notifications	1 : N	One user can receive multiple notifications.
Users	Reports	1 : N	One authorized user can generate multiple reports.
Classes	Reports	1 : N	One class can have multiple reports generated for it.

RELATIONSHIP DIAGRAM
School Timetable and Attendance Management System



4. Sample Data

The sample data below illustrates how records are stored in each table once the system is populated. These records demonstrate the relationships between users, students, teachers, classes, timetable, attendance, and leave modules. The data is provided for illustration purposes only, to help understand the database structure and flow of information within the School Timetable and Attendance Management System.

Users

user_id	name	role
U001	Aarav Sharma	STUDENT
U002	Nisha Reddy	TEACHER
U003	Admin User	ADMIN

Classes

class_id	class_name	section	class_teacher_id
C001	X	A	U002
C002	IX	B	U004

Students

student_id	roll_no	name	class_id
ST001	1	Aarav Sharma	C001
ST002	2	Diya Patel	C001

Teachers

teacher_id	name	subject_specialization
T001	Nisha Reddy	English
T002	Ravi Kumar	Maths

Timetable

timetable_id	class_id	day_of_week	period_number
TT001	C001	MONDAY	1
TT002	C001	MONDAY	2

Attendance

attendance_id	student_id	attendance_date	status
A001	ST001	12/05/2024	PRESENT
A002	ST002	12/05/2024	ABSENT

LeaveApplication

leave_id	applicant_id	leave_type	status
L001	U002	MEDICAL	APPROVED
L002	U004	CASUAL	PENDING